

Network Services Configuration -- Guided Notes ANSWER KEY

Unit 2.5.1 | CompTIA Network+ N10-009 Obj. 2.5 | N10-008 2.5

For instructor use / distribute after students complete the notes.

Question 1

A DHCP server automatically assigns _____, _____, _____, and _____ to clients on a network. Without DHCP, these values would need to be configured _____ on every device.

Answer: IP address; subnet mask; default gateway; DNS server address; manually (statically).

Question 2

The four steps of the DHCP process in order are: _____, _____, _____, _____. The memory aid for this sequence is _____.

Answer: Discover, Offer, Request, Acknowledge (DORA). Memory aid: DORA.

Question 3

A DHCP scope defines the _____ of addresses available for assignment. An exclusion range removes specific addresses from the pool (typically used for _____). A DHCP reservation assigns the _____ IP address to a device based on its MAC address.

Answer: Range (pool); servers, printers, and other devices needing static IPs; same (fixed/permanent).

Question 4

When a client on one subnet needs to obtain a DHCP address from a server on a different subnet, a _____ agent must be configured on the router interface. This agent forwards DHCP _____ packets across subnet boundaries since these packets cannot normally cross a router.

Answer: DHCP relay agent (ip helper-address); broadcast.

Question 5

DNS resolves _____ names (like www.example.com) to _____ addresses. A DNS _____ record maps a hostname to an IPv4 address. A _____ record maps a hostname to an IPv6 address. A _____ record maps an IP address back to a hostname.

Answer: Domain (human-readable); IP; A record; AAAA record; PTR record (reverse lookup).

Question 6

NTP (Network Time Protocol) synchronizes _____ across network devices. Accurate time is critical for _____, _____, and _____. NTP uses a hierarchy of servers called _____ levels, with atomic clocks at stratum _____.

Answer: Clocks (system time); log correlation, security certificates/authentication, and Kerberos/AD; stratum; 0.

Question 7

Explain the difference between a DHCP lease and a DHCP reservation. When would you use each, and what happens when a DHCP lease expires if the client is still connected?

Answer: A lease is a temporary assignment from the address pool -- any available address is assigned for a set duration (e.g., 8 days). A reservation permanently assigns a specific IP to a specific MAC address, used for printers, servers, and cameras. When a lease expires on an active client, the client attempts to renew with a DHCP Request; if the server responds with an Acknowledge, the lease is extended.

Question 8 -- Real World Application

REAL WORLD: Users on a network segment report that some devices receive IP addresses in the expected range (192.168.10.x) but others receive addresses in the 169.254.x.x range. Explain what 169.254.x.x addresses indicate, list three possible causes, and describe the first diagnostic command you would run.

Answer: 169.254.x.x is an APIPA (Automatic Private IP Addressing) address -- the client could not reach a DHCP server and self-assigned a link-local address. Possible causes: (1) DHCP scope is exhausted; (2) DHCP server is down or unreachable; (3) DHCP relay agent is misconfigured or missing. First command: ipconfig /all on an affected client to confirm no DHCP server is listed, then check scope statistics for lease exhaustion.